Marcelo d'Amorim

Associate Professor, Computer Science Department
North Carolina State University

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Education

University of Illinois at Urbana-Champaign ..... Champaign, IL, US
2007 Ph.D. in Computer Science
Dissertation title: "Efficient Explicit-State Model Checking for Programs with Dynamically Allocated Data"
Advisor: Darko Marinov
Federal University of Pernambuco ..... Recife, PE, Brazil
2001 M.S. in Computer Science
1997 B.S. in Computer Science

Research Interests My research interests are in the areas of **Software Engineering and Programming Languages**, with a focus on improving software reliability through program analysis and systematic testing.

Experience

22– **North Carolina State University** ..... Raleigh, NC, US
Associate Professor. Advising: 6 PhDs; Graduated: 1 PhD
09–22 **Federal University of Pernambuco (UFPE)** ..... Recife, PE, Brazil
Associate Professor. Graduated: 2 PhD + 12 MS
15–16 **Georgia Institute of Technology** ..... Atlanta, GA, US
Visiting Scholar at the Arktos group led by Alessandro Orso
07–08 **Federal University of Pernambuco** ..... Recife, PE, Brazil
Postdoctoral researcher at the SPG group led by Paulo Borba

Software https://github.com/ncsu-swat

Service https://damorim.github.io/service.html

Current PhD Students

22- M. M. Abid Naziri [[Qualified thesis proposal.]]
22- Ishrak Hayet [[Passed written exam. Thesis proposal qualification exam scheduled.]]
24- Feiran Qin [[Passed written exam.]]
24- Lingjun Liu [[Scheduled written exam.]]
24- Mohsen EsfandyariDoulabi [[Scheduled written exam.]]
25- Giorgi Merabishvili

Graduated PhD Students

PhD'26 Denini Silva (UFPE), Detection and Prevention of Resource-Aware Flaky Tests
First job: Rural Federal University of Pernambuco (Brazil)
PhD'24 Sarah Elder (Co-advised with Laurie Williams), Supply-Chain Security Analysis with Score Cards
First job: University of North Carolina Willmington
PhD'15 Sabrina Souto, Addressing High Dimensionality and Lack of Feature Models in Testing of Software Product Lines
First job: State University of Paraíba (Brazil)

Graduated MS Students

MS'25 MohamadAli Farahat, non-thesis masters
MS'23 Paulo Nunes, Detecting Failures in Autonomous Driving with Explainable AI
As of Aug'23: Software Architect at Stelantis
MS'23 Beatriz Souza, Automated Detection of Code-Comment Inconsistencies
As of Aug'23: PhD student at University of Stuttgart, Germany
MS'20 Igor Lima, Leveraging Diversity to Find Bugs in JavaScript Engines
As of Aug'22: Software Developer at Sensedia, Brazil
MS'19 Luis Melo, Using Docker to Assist QA Forum Users
As of Aug'22: Software Development Engineer at AWS, Canada

MS'18 Jeanderson Cândido, Test Suite Parallelization in Open-Source Projects
As of Aug'22: PhD student at TU Delft, Netherlands

MS'16 Paulo Barros, Resolving Java Reflection and Android Intents
As of Aug'22: Senior Software Engineer at Chronicled, Brazil

MS'15 Mateus Borges, qCORAL: Quantitative Constraint Solver for Complex Mathematical Constraints
As of Aug'22: Senior Backend Engineer at Elinvar Gmbh, Germany

MS'12 Elton Alves, Improved Fault Localization with Dynamic Slicing and Change Impact Analysis
As of Aug'22: Senior Software Engineer at Zartis, Spain

MS'12 João Paulo Oliveira, Rabbit - A Novel Approach to Find Data Races in Concurrent Programs
Co-advised with Fernando Castor, As of Aug'22: CEO NoxBitcoin, Brazil

MS'10 Andrei Rimsa Alvares, Efficient Static Analysis to Find Tainted Variable Attacks
Co-advised with Fernando Pereira and Roberto Bigonha, As of Aug'22: Assistant Professor at CEFET-MG, Brazil

MS'09 Mitsuo Takaki, Effective CSP solvers with Particle-Swarm Optimization and Genetic Algorithms
Co-advised with Ricardo Prudêncio, As of Aug'22: Principal Software Engineer at Absolute Software, Canada

MS'08 Gláucia Peres, A Black-box Testing Technique to Detect Crashes Based on Automated Test Scenarios
Co-advised with Alexandre Mota, As of Aug'22: Director of Engineering at FreshBooks, Canada

Funding

Table 1 shows the awarded grants since I joined NC State in November 2022. Each row of the table corresponds to one grant. The columns show, from left to right, the sponsor, the period, the PIs, the total amount of the grant, the university share, and my share. The row in gray color shows a grant transferred to me when a colleague left NC State.

Table 1: Awarded grants sorted in reverse chronological order of end date.

Title	Sponsor	Budget Period	PIs / Co-PIs	Total	Univ. Share	My Share
1	NSF	07/01/2024 - 06/30/2027	M. d'Amorim	572,778	572,778	**572,778**
2	NSF	09/01/2023 - 02/28/2026	M. d'Amorim / O. Legunsen	100,000	50,000	**50,000**
3	L3Harris	08/19/2024 - 12/31/2025	T. Menzies / M. d'Amorim	316,000	316,000	**133,365**
4	NSF	10/1/2020 - 09/30/2025	A. Rahman / M. d'Amorim (C. Parnin)	444,720	199,978	**199,978**
Σ				1,433,498	$1,138,756	$956,121

1. SHF: Small: E2R2–A Comprehensive Approach to Improve Simulation-based Testing of Autonomous Driving Systems
2. Collaborative Research: FMitF: Track II: Cross-Language Support for Runtime Verification
3. Software Defect Discovery
4. Collaborative Research: SaTC: TTP: Small: eSLIC: Enhanced Security Static Analysis for Detecting Insecure Configuration Scripts

Teaching

Classes I taught at NC State. The letters U and G indicate **U**ndergraduate- and **G**raduate-level courses.

Term	Class	Level	Enrollment
Sp 2026	CSC 510: Software Engineering	G	53
Fa 2025	CSC 712: Software Testing and Reliability	G	21
Sp 2025	CSC 495: Special Topics in Software Testing	U	12
Fa 2024	CSC 591/712: Software Testing and Reliability	G	41 (34/7)
Sp 2024	**No teaching due to FSE'24 organization**		
Fa 2023	CSC 712: Software Testing and Reliability	G	11
Sp 2023	CSC 510: Software Engineering	G	60

Classes I taught at the Federal University of Pernambuco (UFPE) in reverse-chronological order since I joined in 2009 until I left in 2022. The academic year at UFPE consists of two semesters, referred to as year.1 and year.2.

Term	Class 1	Class 2
2022.1	Software Engineering (U), 53 students	Advanced Software Testing (G), 22 students
2021.2	Software Engineering (U), 48 students	Advanced Software Testing (G), 16 students
2021.1	Software Engineering (U), 58 students	Advanced Software Testing (G), 19 students
2020.2	Compilers (U), 22 students	Software Testing (U), 27 students
2020.1	Compilers (U), 62 students	Software Testing (U), 17 students
2019.2	Compilers (U), 24 students	Software Testing (U), 34 students

2019.1	Compilers (U), 43 students	Advanced Software Testing (G), 21 students
2018.2	Compilers (U), 14 students	Advanced Software Testing (G), 15 students
2018.1	Compilers (U), 27 students	Advanced Software Testing (G), 18 students
2017.2	Compilers (U), 21 students	Advanced Software Testing (G), 34 students
2017.1	Compilers (U), 22 students	Advanced Software Testing (G), 20 students
2016.2	Compilers (U), 29 students	Advanced Software Testing (G), 28 students
2016.1 2015.2	sabbatical	
2015.1	Compilers (U), 45 students	Seminar in Software Testing (G), 18 students
2014.2	Compilers (U), 34 students	Seminar in Software Testing (G), 7 students
2014.1	Compilers (U), 35 students	Seminar in Software Testing (G), 9 students
2013.2	Compilers (U), 40 students	Seminar in Software Testing (G), 9 students
2013.1	Compilers (U), 31 students	Seminar in Software Testing (G), 12 students
2012.2	Compilers (U), 32 students	Introduction to Computing (U), 68 students
2012.1	Compilers (U), 38 students	Introduction to Computing (U), 71 students
2011.2	Compilers (U), 40 students	Introduction to Computing (U), 73 students
2011.1	Compilers (U), 38 students	Introduction to Computing (U), 70 students
2010.2	Introduction to Computing (U), 72 students	Introduction to Computing (U), 80 students
2010.1	Functional Programming (U), 15 students	Introduction to Static Analysis (G), 10 students
2009.2	Operating Systems (U), 24 students	Introduction to Computing (U), 91 students
2009.1	Introduction to Computing (U), 77 students	Introduction to Computing (U), 67 students

Refereed Conference papers

[C1] F. Molina, M. M. A. Naziri, F. Qin, A. Gorla, and M. d'Amorim, "Improving deep learning library testing with machine learning," in *Proceedings of the ACM/IEEE International Conference on Automation of Software Test (AST)*, 2026.

[C2] F. Qin, M. M. A. Naziri, H. Ai, S. Dutta, and M. d'Amorim, "How effective is coverage-guided fuzzing to test deep learning library apis?" In *Proceedings of the IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2026.

[C3] P. Jiang, K. Guan, M. Khosravi, M. Ismail, M. d'Amorim, and O. Legunsen, "Fine-grained analyses for evolution-aware runtime verification," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C4] M. M. A. Naziri, S. C. Lambertenghi, A. Stocco, and M. d'Amorim, "Misbehavior forecasting for focused autonomous driving systems testing," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C5] M. M. A. Naziri, S. Kim, F. Qin, S. Dutta, and M. d'Amorim, "Testing deep learning libraries via neurosymbolic constraint learning," in *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026.

[C6] D. Silva, M. Farahat, and M. d'Amorim, "An empirical analysis of cross-os portability issues in python projects," in *Proceedings of the International Conference on Mining Software Repositories (MSR)*, 2026.

[C7] L. Liu, F. Qin, O. Legunsen, and M. d'Amorim, "Learning compiler fuzzing mutators from historical bugs," in *Proceedings of the International Conference on Mining Software Repositories (MSR)*, 2026.

[C8] K. Guan, M. d'Amorim, et al., "Faster explicit-trace monitoring-oriented programming for runtime verification of software tests," in *Proceedings of the ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2025.

[C9] U. Kulsum, H. Zhu, B. Xu, and M. d'Amorim, "A case study of LLM for automated vulnerability repair: Assessing impact of reasoning and patch validation feedback," in *Proceedings of the 1st ACM International Conference on AI-Powered Software, AIware 2024, Porto de Galinhas, Brazil, July 15-16, 2024*, 2024.

[C10] I. Hayet, A. Scott, and M. d'Amorim, "Feedback-directed partial execution," in *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis, ISSTA 2024, Vienna, Austria, September 16-20, 2024*, 2024, pp. 781–793.

[C11] R. Lima, K. Costa, J. Souza, L. Teixeira, B. Fonseca, M. d'Amorim, M. Ribeiro, and B. Miranda, "Do you see any problem? on the developers perceptions in test smells detection," in *Proceedings of the XXII Brazilian Symposium on Software Quality, SBQS 2023, Brasilia, Brazil, November 7-10, 2023*, 2023, pp. 21–30.

Highlighted text indicates papers published while at North Carolina State University.

[C12] A. Stocco, P. J. Nunes, M. d'Amorim, and P. Tonella, "Thirdeye: Attention maps for safe autonomous driving systems," in *37th IEEE/ACM International Conference on Automated Software Engineering, ASE 2022, Rochester, MI, USA, October 10-14, 2022*, 2022, 102:1–102:12.

[C13] S. Reis, R. Abreu, M. d'Amorim, and D. Fortunato, "Leveraging practitioners' feedback to improve a security linter," in *37th IEEE/ACM International Conference on Automated Software Engineering, ASE 2022, Rochester, MI, USA, October 10-14, 2022*, 2022, 66:1–66:12.

[C14] F. Molina, M. d'Amorim, and N. Aguirre, "Fuzzing class specifications," in *44th IEEE/ACM 44th International Conference on Software Engineering, ICSE 2022, Pittsburgh, PA, USA, May 25-27, 2022*, 2022, pp. 1008–1020.

[C15] L. Teixeira, B. Miranda, H. Rebêlo, and M. d'Amorim, "Demystifying the challenges of formally specifying API properties for runtime verification," in *14th IEEE Conference on Software Testing, Verification and Validation, ICST 2021, Porto de Galinhas, Brazil, April 12-16, 2021*, 2021, pp. 82–93.

[C16] J. Henkel, D. Silva, L. Teixeira, M. d'Amorim, and T. W. Reps, "Shipwright: A human-in-the-loop system for dockerfile repair," in *43rd IEEE/ACM International Conference on Software Engineering, ICSE 2021, Madrid, Spain, 22-30 May 2021*, 2021, pp. 1148–1160.

[C17] S. Mondal, D. Silva, and M. d'Amorim, "Soundy automated parallelization of test execution," in *IEEE International Conference on Software Maintenance and Evolution, ICSME 2021, Luxembourg, September 27 - October 1, 2021*, 2021, pp. 309–319.

[C18] B. Miranda, I. Lima, O. Legunsen, and M. d'Amorim, "Prioritizing runtime verification violations," in *13th IEEE International Conference on Software Testing, Validation and Verification, ICST 2020, Porto, Portugal, October 24-28, 2020*, 2020, pp. 297–308.

[C19] M. A. Guimarães, L. Fernandes, M. Ribeiro, M. d'Amorim, and R. Gheyi, "Optimizing mutation testing by discovering dynamic mutant subsumption relations," in *13th IEEE International Conference on Software Testing, Validation and Verification, ICST 2020, Porto, Portugal, October 24-28, 2020*, 2020, pp. 198–208.

[C20] D. Silva, L. Teixeira, and M. d'Amorim, "Shake it! detecting flaky tests caused by concurrency with shaker," in *IEEE International Conference on Software Maintenance and Evolution, ICSME 2020, Adelaide, Australia, September 28 - October 2, 2020*, 2020, pp. 301–311.

[C21] G. Pinto, B. Miranda, S. Dissanayake, M. d'Amorim, C. Treude, and A. Bertolino, "What is the vocabulary of flaky tests?" In *MSR '20: 17th International Conference on Mining Software Repositories, Seoul, Republic of Korea, 29-30 June, 2020*, 2020, pp. 492–502.

[C22] X. Li, M. d'Amorim, and A. Orso, "Intent-preserving test repair," in *12th IEEE Conference on Software Testing, Validation and Verification, ICST 2019, Xi'an, China, April 22-27, 2019*, 2019, pp. 217–227.

[C23] S. Reis, R. Abreu, and M. d'Amorim, "Demystifying the combination of dynamic slicing and spectrum-based fault localization," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2019.

[C24] M. Fazzini, M. Prammer, M. d'Amorim, and A. Orso, "Automatically Translating Bug Reports into Test Cases for Mobile Apps," in *Proceedings of the 27th ACM SIGSOFT International Symposium on Software Testing and Analysis, ISSTA 2018, Amsterdam, The Netherlands, July 16-21, 2018*, 2018, pp. 141–152.

[C25] X. Li, S. Zhu, M. d'Amorim, and A. Orso, "Enlightened Debugging," in *Proceedings of the 40th International Conference on Software Engineering, ICSE 2018, Gothenburg, Sweden, May 27 - June 03, 2018*, 2018, pp. 82–92.

[C26] A. Perez, R. Abreu, and M. D'Amorim, "Prevalence of single-fault fixes and its impact on fault localization," in *2017 IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2017, pp. 12–22.

[C27] J. Candido, L. Melo, and M. d'Amorim, "Test suite parallelization in open-source projects: A study on its usage and impact," in *Proceedings of the 32nd IEEE/ACM International Conference on Automated Software Engineering, ASE 2017, Urbana, IL, USA, October 30 - November 03, 2017*, 2017, pp. 838–848.

[C28] S. Souto, M. d'Amorim, and R. Gheyi, "Balancing soundness and efficiency for practical testing of configurable systems," in *Proceedings of the 39th International Conference on Software Engineering, ICSE 2017, Buenos Aires, Argentina, May 20-28, 2017*, 2017, pp. 632–642.

[C29] X. Li, M. d'Amorim, and A. Orso, "Iterative user-driven fault localization," in *Hardware and Software: Verification and Testing - 12th International Haifa Verification Conference, HVC 2016, Haifa, Israel, November 14-17, 2016, Proceedings*, 2016, pp. 82–98.

[C30] M. Borges, A. Filieri, M. d'Amorim, and C. S. Păsăreanu, "Iterative distribution-aware sampling for probabilistic symbolic execution," in *Proceedings of the 10th Joint Meeting on Foundations of Software Engineering*, 2015, pp. 866–877.

[C31] P. Barros, R. Just, S. Millstein, P. Vines, W. Dietl, M. d'Amorim, and M. D. Ernst, "Static Analysis of Implicit Control Flow: Resolving Java Reflection and Android Intents," in *Proceedings of the 30th IEEE/ACM International Conference on Automated Software Engineering*, 2015, pp. 669–679.

[C32] T. Liu, M. Araújo, M. d'Amorim, and M. Taghdiri, "A comparative study of incremental constraint solving approaches in symbolic execution," in *Proceedings of the 10th International Haifa Verification Conference*, 2014, pp. 284–299.

[C33] M. Borges, A. Filieri, M. d'Amorim, C. S. Păsăreanu, and W. Visser, "Compositional solution space quantification for probabilistic software analysis," in *Proceedings of the 35th ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2014, pp. 123–132.

[C34] J. Campos, R. Abreu, G. Fraser, and M. d'Amorim, "Entropy-based test generation for improved fault localization," in *Proceedings of the 28th IEEE/ACM International Conference on Automated Software Engineering*, 2013, pp. 257–267.

[C35] C. H. P. Kim, D. Marinov, S. Khurshid, D. Batory, S. Souto, P. Barros, and M. d'Amorim, "SPLat: Lightweight Dynamic Analysis for Reducing Combinatorics in Testing Configurable Systems," in *Proceedings of the 9th Joint Meeting on Foundations of Software Engineering*, 2013, pp. 257–267.

[C36] M. Borges, M. d'Amorim, S. Anand, D. H. Bushnell, and C. S. Pasareanu, "Symbolic execution with interval solving and meta-heuristic search," in *Proceedings of the Fifth IEEE International Conference on Software Testing, Verification and Validation*, 2012, pp. 111–120.

[C37] E. Alves, M. Gligoric, V. Jagannath, and M. d'Amorim, "Fault-localization using dynamic slicing and change impact analysis," in *26th IEEE/ACM International Conference on Automated Software Engineering (ASE 2011), Lawrence, KS, USA, November 6-10, 2011*, 2011, pp. 520–523.

[C38] M. Souza, M. Borges, M. d'Amorim, and C. S. Pasareanu, "CORAL: solving complex constraints for symbolic pathfinder," in *NASA Formal Methods - Third International Symposium, NFM 2011, Pasadena, CA, USA, April 18-20, 2011. Proceedings*, 2011, pp. 359–374.

[C39] A. Rimsa, M. d'Amorim, and F. M. Q. Pereira, "Tainted flow analysis on e-ssa-form programs," in *Proceedings of the 20th International Conference on Compiler Construction: Part of the Joint European Conferences on Theory and Practice of Software*, 2011, pp. 124–143.

[C40] M. Takaki, D. Cavalcanti, R. Gheyi, J. Iyoda, M. d'Amorim, and R. B. C. Prudêncio, "A comparative study of randomized constraint solvers for random-symbolic testing," in *First NASA Formal Methods Symposium - NFM 2009, Moffett Field, California, USA, April 6-8, 2009.*, 2009, pp. 56–65.

[C41] C. Bertolini, G. Peres, M. d'Amorim, and A. Mota, "An empirical evaluation of automated black box testing techniques for crashing guis," in *Second International Conference on Software Testing Verification and Validation, ICST 2009, Denver, Colorado, USA, April 1-4, 2009*, 2009, pp. 21–30.

[C42] M. d'Amorim, S. Lauterburg, and D. Marinov, "Delta execution for efficient state-space exploration of object-oriented programs," in *Proceedings of the 2007 International Symposium on Software Testing and Analysis*, 2007, pp. 50–60.

[C43] F. Chen, M. d'Amorim, and G. Roşu, "Checking and correcting behaviors of java programs at runtime with java-mop," *Electronic Notes in Theoretical Computer Science*, vol. 144, no. 4, pp. 3–20, 2006.

[C44] M. d'Amorim, C. Pacheco, T. Xie, D. Marinov, and M. D. Ernst, "An empirical comparison of automated generation and classification techniques for object-oriented unit testing," in *Proceedings of the 21st IEEE/ACM International Conference on Automated Software Engineering*, 2006, pp. 59–68.

[C45] M. d'Amorim, A. Sobeih, and D. Marinov, "Optimized execution of deterministic blocks in java pathfinder," in *Proceedings of the 8th International Conference on Formal Methods and Software Engineering*, 2006, pp. 549–567.

[C46] M. d'Amorim and G. Roşu, "Efficient monitoring of ω-languages," in *Proceedings of 17th International Conference on Computer-aided Verification*, vol. 3576, 2005, pp. 364–378.

[C47] F. Chen, M. d'Amorim, and G. Rosu, "A formal monitoring-based framework for software development and analysis," in *Formal Methods and Software Engineering, 6th International Conference on Formal Engineering Methods, ICFEM 2004, Seattle, WA, USA, November 8-12, 2004, Proceedings*, vol. 3308, 2004, pp. 357–372.

Journals

[J1] X. Chen, M. Biagiola, V. Riccio, M. d'Amorim, and A. Stocco, "XMutant: XAI-based fuzzing for deep learning systems," *Empirical Software Engineering*, vol. 31, no. 4, p. 102, 2026.

[J2] R. Melo, P. Simões, R. Gheyi, M. d'Amorim, M. Ribeiro, G. Soares, E. Almeida, and E. Soares, "Agentic lms: Hunting down test smells," *IEEE Software*, vol. 43, no. 1, pp. 32–40, 2026.

[J3] Y. Zhang, U. Paul, M. d'Amorim, and A. Rahman, "Configuration defects in kubernetes," *IEEE Transactions on Software Engineering*, vol. 52, no. 3, pp. 1125–1144, 2026.

Highlighted text indicates papers published while at North Carolina State University.

[J4] I. Hayet, A. Scott, and M. d'Amorim, "ChatAssert: LLM-based Test Oracle Generation with External Tools Assistance," *IEEE Transactions on Software Engineering*, no. 01, pp. 1–18, 2025.

[J5] F. Molina and A. Gorla, "Test oracle automation in the era of LLMs," *ACM Transactions on Software Engineering and Methodology*, 2025.

[J6] L. Pasquale, A. Sabetta, M. d'Amorim, P. Hegedüs, M. T. Mirakhorli, H. Okhravi, M. Payer, A. Rashid, J. C. S. Santos, J. M. Spring, L. Tan, and K. Tuma, "Challenges to using large language models in code generation and repair," *IEEE Security & Privacy*, vol. 23, no. 2, pp. 81–88, 2025.

[J7] D. Silva, M. Gruber, S. Gokhale, E. Arteca, A. Turcotte, M. d'Amorim, W. Lam, S. Winter, and J. Bell, "The Effects of Computational Resources on Flaky Tests," *IEEE Transactions on Software Engineering*, no. 01, pp. 1–18, 2024.

[J8] K. Barbosa, R. Ferreira, G. Pinto, M. d'Amorim, and B. Miranda, "Test flakiness across programming languages," *IEEE Trans. Software Eng.*, vol. 49, no. 4, pp. 2039–2052, 2023.

[J9] B. Reid, M. d'Amorim, M. Wagner, and C. Treude, "NCQ: code reuse support for node.js developers," *IEEE Trans. Software Eng.*, vol. 49, no. 5, pp. 3205–3225, 2023.

[J10] A. Torres, P. H. T. Costa, L. H. V. Amaral, J. Pastro, R. Bonifácio, M. d'Amorim, O. Legunsen, E. Bodden, and E. D. Canedo, "Runtime verification of crypto apis: An empirical study," *IEEE Trans. Software Eng.*, vol. 49, no. 10, pp. 4510–4525, 2023.

[J11] L. Alcantara, G. Padilha, R. Abreu, and M. d'Amorim, "Syrius: Synthesis of rules for intrusion detectors," *IEEE Trans. Reliab.*, vol. 71, no. 1, pp. 370–381, 2022.

[J12] L. Cabral, B. Miranda, I. Lima, and M. d'Amorim, "Rvprio: A tool for prioritizing runtime verification violations," *Softw. Test. Verification Reliab.*, vol. 32, no. 5, 2022.

[J13] R. Gheyi, M. Ribeiro, B. Souza, M. A. Guimarães, L. Fernandes, M. d'Amorim, V. Alves, L. Teixeira, and B. Fonseca, "Identifying method-level mutation subsumption relations using Z3," *Inf. Softw. Technol.*, vol. 132, p. 106 496, 2021.

[J14] I. Lima, J. Silva, B. Miranda, G. Pinto, and M. d'Amorim, "Exposing bugs in javascript engines through test transplantation and differential testing," *Softw. Qual. J.*, vol. 29, no. 1, pp. 129–158, 2021.

[J15] L. Melo, I. Wiese, and M. d'Amorim, "Using docker to assist q&a forum users," *IEEE Trans. Software Eng.*, vol. 47, no. 11, pp. 2563–2574, 2021.

[J16] I. Lima, J. Cândido, and M. d'Amorim, "Practical detection of CMS plugin conflicts in large plugin sets," *Inf. Softw. Technol.*, vol. 118, 2020.

[J17] S. Souto and M. d'Amorim, "Time-space efficient regression testing for configurable systems," *Journal of Systems and Software*, vol. 137, pp. 733–746, 2018.

[J18] A. Rimsa, M. d'Amorim, F. M. Q. Pereira, and R. S. Bigonha, "Efficient static checker for tainted variable attacks," *Science of Computer Programming*, vol. 80, pp. 91–105, 2014.

[J19] A. Sobeih, M. d'Amorim, M. Viswanathan, D. Marinov, and J. C. Hou, "Assertion checking in j-sim simulation models of network protocols," *Simulation*, vol. 86, no. 11, pp. 651–673, 2010.

[J20] M. Takaki, D. Cavalcanti, R. Gheyi, J. Iyoda, M. d'Amorim, and R. B. C. Prudêncio, "Randomized constraint solvers: A comparative study," *Innovations in Systems and Software Engineering*, vol. 6, no. 3, pp. 243–253, 2010.

[J21] M. d'Amorim, S. Lauterburg, and D. Marinov, "Delta execution for efficient state-space exploration of object-oriented programs," *IEEE Transactions on Software Engineering*, vol. 34, no. 5, pp. 597–613, 2008.

[J22] F. Chen, M. d'Amorim, and G. Roşu, "Checking and correcting behaviors of java programs at runtime with java-mop," *Electronic Notes in Theoretical Computer Science*, vol. 144, no. 4, pp. 3–20, 2006.

[J23] M. d'Amorim and K. Havelund, "Event-based runtime verification of java programs," *SIGSOFT Software Engineering Notes*, vol. 30, no. 4, pp. 1–7, 2005.

Short Papers (Conference and Workshop)

[S1] Z. Shen, M. Yaseen, K. Guan, D. Silva, M. d'Amorim, and O. Legunsen, "Pymop: A runtime verification tool for python," in *Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE): Tool Demonstrations*, 2026.

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Highlighted text indicates papers published while at North Carolina State University.

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